Comparative Analysis of Interview-Based and Social Media-Based Radicalisation Indicators: A Cross-Platform Validation Study for Indonesian Extremism Detection

Executive Summary

This 12-month research project conducts a systematic comparative analysis between radicalisation indicators derived from in-depth interviews (N=200+ convicts/ex-convicts) and those expressed in social media contexts (100 Facebook profiles). Building on our existing AI model trained on MacAdams interview data using 3N Theory and Identity Fusion indicators, this study validates and extends the framework to social media environments. The project will establish the transferability, limitations, and platform-specific variations of radicalisation indicators, ultimately creating a unified cross-platform detection system with 70-80% classification accuracy.

Chapter 1: Introduction and Research Context

Background and Rationale

Radicalisation detection has evolved from traditional interview-based assessments to digital surveillance approaches. However, a critical gap exists in understanding how radicalisation indicators manifest differently across communication platforms. This research addresses this gap by conducting the first systematic comparison between interview-derived and social media-derived radicalisation indicators in the Indonesian context.

Research Questions

RQ1: To what extent do radicalisation indicators identified through in-depth interviews (MacAdams protocol) align with or diverge from indicators expressed in naturalistic social media environments?

RQ2: How do platform-specific factors (anonymity, audience, temporal dynamics) influence the expression and detectability of radicalisation indicators?

RQ3: Can AI models trained on interview data effectively transfer to social media contexts, and what adaptations are required for optimal performance?

RQ4: What is the optimal integration strategy for combining interview-based and social media-based indicators into a unified assessment framework?

Innovation and Contribution

This research advances the field by:

- **First systematic cross-platform validation** of radicalisation indicators
- **Evidence-based framework** for adapting interview protocols to digital contexts

- **Validated methodology** for transferring psychological assessment tools to social media analysis
- **Unified detection system** leveraging both traditional and digital assessment approaches

Existing Foundation

This project builds upon our completed work:

- **Codebook Development:** Comprehensive indicator framework based on 3N Theory (Need, Narrative, Network) and Identity Fusion Theory
- **Interview Dataset:** 200+ MacAdams life-story interviews with Indonesian convicts and ex-convicts
- **Baseline AI Model:** Trained classification system achieving 75% accuracy on interview transcripts
- **Validated Indicators:** 42 empirically-derived indicators spanning person-centric, psychosocial, and protective factors

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Chapter 2: Literature Review

2.1 Theoretical Foundations of Radicalisation

2.1.1 Significance Quest Theory (Kruglanski et al., 2014, 2019)

The quest for personal significance drives radicalisation through three components:

- **Need:** Loss of significance triggering compensatory mechanisms
- **Narrative:** Ideological frameworks promising significance restoration
- **Network:** Social validation and reinforcement of radical beliefs

Recent empirical validation (Webber et al., 2018; Jasko et al., 2020) demonstrates cross-cultural applicability, though social media manifestations remain understudied.

2.1.2 Identity Fusion Theory (Swann et al., 2012; Whitehouse, 2018)

Identity fusion with extremist groups predicts:

- Willingness to engage in extreme pro-group behavior
- Resistance to de-radicalisation efforts
- Amplification through shared traumatic experiences

Social media research (Bélanger et al., 2020) suggests online fusion mechanisms differ from face-to-face contexts, requiring platform-specific measurement approaches.

2.1.3 Indonesian Context-Specific Models

- **Neo Loo Seng's Framework (2018):** 42 indicators validated on Indonesian extremists

- **BNPT Risk Assessment (2020):** Operational framework used by Indonesian authorities
- **Cultural Factors:** Role of pesantren networks, localized jihadi narratives (Jones, 2018)

2.2 Interview-Based Radicalisation Assessment

2.2.1 MacAdams Life Story Interview Protocol

The MacAdams approach (MacAdams, 2008) elicits narrative identity through:

- Structured prompts about life chapters, turning points, and future scripts
- Coding for redemption/contamination sequences
- Identity development trajectories

Applications to extremism (Hamid, 2020; Atran, 2021) reveal:

- Contamination narratives preceding radicalisation
- Redemptive violence themes in terrorist autobiographies
- Network fusion replacing family bonds

2.2.2 Limitations of Interview Methods

- **Social desirability bias** in post-conviction interviews (Altier et al., 2017)
- **Retrospective reconstruction** affecting accuracy (Horgan, 2012)
- **Limited sample access** constraining generalizability (Nilsson, 2018)

2.3 Social Media-Based Detection Approaches

2.3.1 Computational Methods

Text-Based Approaches:

- Sentiment analysis detecting emotional escalation (Rowe & Saif, 2016)
- Topic modeling identifying ideological themes (Scrivens et al., 2020)
- Linguistic markers of extremism (Cohen et al., 2014)

Network Analysis:

- Homophily patterns in extremist networks (Bermingham et al., 2009)
- Information cascade dynamics (Del Vicario et al., 2016)
- Echo chamber quantification (Cinelli et al., 2021)

2.3.2 AI/ML Applications

Supervised Learning:

- SVM classifiers for extremist content (Agarwal & Sureka, 2015)
- Deep learning for multimodal analysis (Nouh et al., 2019)
- Transformer models for contextual understanding (Alshemali & Kalita, 2020)

Performance Benchmarks:

- Binary classification (extremist/non-extremist): 85-92% accuracy
- Multi-class severity: 70-78% accuracy
- Cross-platform transfer: 15-20% performance degradation

2.3.3 Cross-Platform Validation Gap

Limited research exists comparing:

- Interview vs. social media indicator expression
- Offline vs. online radicalisation trajectories
- Traditional assessment vs. computational detection validity

2.4 Explainable AI in Security Contexts

2.4.1 Interpretability Requirements

- Legal standards for automated decision-making (GDPR Article 22)
- Operational need for actionable intelligence
- Trust and adoption by security practitioners

2.4.2 Technical Approaches

- Attention mechanisms for highlighting relevant text (Mohseni et al., 2021)
- LIME/SHAP for feature importance (Ribeiro et al., 2016)
- Concept activation vectors for high-level explanations (Kim et al., 2018)

2.5 Research Gap Analysis

Current literature reveals critical gaps:

1. **No systematic comparison** between interview and social media indicators
2. **Limited cross-platform validation** of radicalisation theories
3. **Absence of integrated frameworks** combining traditional and computational approaches
4. **Insufficient attention** to platform-specific expression patterns
5. **Lack of Indonesian-specific** social media radicalisation studies

This research directly addresses these gaps through comparative analysis and integrated model development.

Chapter 3: Methodology

3.1 Research Design Overview

Mixed-Methods Comparative Design:

- Qualitative comparison of indicator manifestations across platforms
- Quantitative validation of model transferability
- Integration through joint codebook development

3.2 Phase 1: Baseline Establishment and Data Preparation (Months 1-3)

3.2.1 Existing Interview-Based Foundation

Available Resources:

- 200+ coded MacAdams interviews (completed)
- Validated codebook with 3N and Identity Fusion indicators
- Trained AI model (75% accuracy baseline)

Preparation Tasks:

- Codebook documentation and standardization
- Model performance benchmarking on interview data
- Indicator frequency and reliability analysis

3.2.2 Social Media Data Collection

Sampling Strategy:

- 100 Indonesian Facebook profiles
- Stratified by conviction status (convicted terrorists, supporters, sympathizers)
- Minimum 1-year posting history per profile
- Balanced representation across extremist groups (JI, JAD, MIT)

Data Processing:

- Secure API-based collection adhering to platform ToS
- Indonesian-to-English translation with back-translation validation
- Metadata extraction (timestamps, engagement, network features)
- Privacy protection through immediate anonymization

3.3 Phase 2: Comparative Codebook Development (Months 3-5)

3.3.1 Initial Mapping Exercise

Interview → Social Media Translation:

- Systematic review of each interview-based indicator
- Identification of potential social media manifestations
- Documentation of platform-specific constraints

Example Mapping:

Interview Indicator	Social Media Manifestation	Platform Considerations
"Significance loss narrative"	Status updates about failures, grievances	Public vs. private sharing settings
"Network fusion"	Group membership, interaction patterns	Algorithmic amplification effects
"Redemptive violence"	Sharing martyrdom content, glorification	Content moderation circumvention

3.3.2 Emergent Indicator Discovery

Inductive Analysis:

- Open coding of social media posts not captured by interview codebook
- Identification of platform-unique indicators
- LLM-assisted pattern recognition for scalability

Validation Process:

- Expert panel review (psychology, security, Indonesian culture)
- Inter-rater reliability testing (Krippendorff's $\alpha \geq 0.80$)
- Iterative refinement based on disagreements

3.4 Phase 3: Dual-Platform Annotation (Months 5-7)

3.4.1 Annotation Protocol

Multi-Level Coding:

1. **Post Level:** Individual posts coded for risk and indicators
2. **Profile Level:** Aggregate assessment across posting history
3. **Temporal Level:** Trajectory analysis over time

Annotation Team:

- 3 trained coders familiar with interview codebook
- 2 social media specialists

- 1 Indonesian cultural expert

3.4.2 Comparative Analysis Framework

Concordance Metrics:

- Indicator co-occurrence patterns
- Severity alignment across platforms
- Temporal expression differences

Divergence Analysis:

- Platform-unique indicators
- Expression intensity variations
- Context-dependent manifestations

3.5 Phase 4: Cross-Platform Model Development (Months 7-9)

3.5.1 Transfer Learning Experiments

Baseline Testing:

- Apply interview-trained model to social media data
- Document performance degradation
- Error analysis by indicator type

Adaptation Strategies:

1. **Fine-tuning:** Adjust interview model with social media data
2. **Domain Adaptation:** Learn platform-invariant representations
3. **Ensemble Approach:** Combine platform-specific models

3.5.2 Unified Model Architecture

Multi-Input Design:

```
Interview Features → Encoder →
                                → Fusion Layer → Risk Classification
Social Media Features → Encoder →
```

Training Strategy:

- Joint optimization on both data types
- Platform-specific dropout for robustness

- Indicator-level multi-task learning

3.6 Phase 5: Validation and Integration (Months 9-12)

3.6.1 Comprehensive Evaluation

Within-Platform Performance:

- Interview model on interview test set
- Social media model on social media test set
- Unified model on both test sets

Cross-Platform Transfer:

- Interview model → social media data
- Social media model → interview data
- Performance degradation analysis

3.6.2 External Validation

Expert Assessment:

- Indonesian security practitioners evaluate outputs
- Clinical psychologists assess indicator validity
- Operational feedback on utility

Comparative Validation:

- Against existing tools (VERA-2R, ERG22+)
- ROC curve analysis for risk thresholds
- Cost-benefit analysis for deployment

Chapter 4: Expected Results and Analysis Plan

4.1 Anticipated Findings

4.1.1 Indicator Concordance

High Concordance Expected:

- Core psychological needs (significance quest)
- Ideological commitment markers
- Network affiliation patterns

Low Concordance Expected:

- Emotional regulation strategies
- Interpersonal conflict patterns
- Childhood trauma narratives

4.1.2 Platform-Specific Patterns

Social Media Unique:

- Viral content amplification
- Echo chamber reinforcement
- Performative extremism

Interview Unique:

- Deep psychological insights
- Contextual life history
- Reflective self-assessment

4.2 Analysis Framework

4.2.1 Quantitative Analyses

Model Performance:

- Classification accuracy by platform
- Precision/recall by risk level
- F1 scores by indicator type

Statistical Comparisons:

- McNemar's test for model differences
- Cohen's kappa for indicator agreement
- Factor analysis for dimensional structure

4.2.2 Qualitative Analyses

Thematic Comparison:

- Narrative structure differences
- Emotional expression variations
- Identity construction patterns

Case Study Analysis:

- Matched pairs (same individual, both platforms)

- Trajectory comparison over time
- Intervention point identification

Chapter 5: Ethical Considerations and Risk Management

5.1 Enhanced Ethical Framework

5.1.1 Dual-Platform Challenges

- Consent complexity for retrospective interview use
- Social media data sensitivity
- Cross-platform identity protection

5.1.2 Algorithmic Fairness

Bias Sources:

- Interview selection bias (only caught extremists)
- Social media self-selection
- Translation quality variations

Mitigation Strategies:

- Stratified evaluation across groups
- Counterfactual fairness testing
- Regular audit protocols

5.2 Operational Ethics

5.2.1 Deployment Considerations

- False positive impact assessment
- Human-in-the-loop requirements
- Appeals process design

5.2.2 Stakeholder Engagement

- Community consultation process
- Civil liberties organization input
- Transparent governance structure

Chapter 6: Project Timeline and Deliverables

6.1 Revised Timeline

Months 1-3: Foundation and Preparation

- Month 1: IRB amendments, social media data collection setup
- Month 2: Interview codebook standardization, team training
- Month 3: Initial social media data collection and processing

Months 3-5: Comparative Analysis

- Month 4: Codebook mapping and adaptation
- Month 5: Pilot annotation and reliability testing

Months 5-7: Full Annotation

- Month 6: Dual-platform annotation implementation
- Month 7: Annotation completion and quality assurance

Months 7-9: Model Development

- Month 8: Transfer learning experiments
- Month 9: Unified model optimization

Months 9-12: Validation and Delivery

- Month 10: Comprehensive evaluation
- Month 11: External validation and stakeholder testing
- Month 12: Final integration and documentation

6.2 Key Deliverables

Academic Outputs:

1. Journal article on cross-platform indicator validation
2. Conference paper on transfer learning for radicalisation detection
3. Methodological piece on comparative codebook development
4. Dataset release (anonymized, with ethics approval)

Practical Tools:

1. Unified assessment system with API
2. Platform-specific adaptation guidelines
3. Training materials for practitioners
4. Evaluation dashboard for continuous monitoring

Chapter 7: Budget and Resources

7.1 Leveraging Existing Resources

Already Available:

- Interview dataset and codebook
- Baseline AI model
- Core research team

Additional Requirements:

- Social media data collection tools
- Translation services
- Extended compute resources
- External validation panel

7.2 Resource Optimization

Cost-Saving Strategies:

- Build on existing model architecture
- Utilize established annotation team
- Leverage partner institution resources

Conclusion

This research plan positions the project as a critical advancement in radicalisation detection by bridging traditional psychological assessment and computational social media analysis. By systematically comparing interview-based and social media-based indicators, we will establish the first evidence-based framework for cross-platform radicalisation assessment.

The comparative approach not only validates existing tools but also identifies platform-specific adaptations necessary for accurate detection. The resulting unified system will provide practitioners with scientifically-grounded tools that leverage both the depth of clinical interviews and the scale of social media analysis.

Success metrics include both technical performance (70-80% classification accuracy) and practical utility (stakeholder adoption and operational integration). The project's emphasis on explainability and ethical deployment ensures responsible advancement of security technologies while respecting civil liberties and cultural contexts.